

Here is your ultimate A-to-Z Master Cheat Sheet. This covers every step of drive provisioning, encryption, hardware-specific optimization, network sharing, and emergency recovery, perfectly tuned for your Raspberry Pi and Docker ecosystem.



Master Cheat Sheet: External Storage Provisioning & Maintenance

Phase 1: Drive Identification & LUKS Encryption

Always format and encrypt drives manually to prevent OS partition leaks.

1. Identify the Drive

```
Bash
lsblk
# Note the target drive, e.g., /dev/sdX1
```

2. Generate a Secure 64-Bit Native Key

```
Bash
sudo mkdir -p /root/.keys
sudo chmod 700 /root/.keys
sudo dd if=/dev/urandom of=/root/.keys/newdrive.key bs=64 count=1
sudo chmod 600 /root/.keys/newdrive.key
```

3. Format and Open the LUKS Vault

```
Bash
sudo cryptsetup luksFormat --type luks2 -d /root/.keys/newdrive.key /dev/sdX1
```

```
sudo cryptsetup luksOpen -d /root/.keys/newdrive.key /dev/sdX1 newdrive_vault
```

4. Create the ext4 Filesystem

```
Bash  
sudo mkfs.ext4 /dev/mapper/newdrive_vault
```

Phase 2: Persistent UUID Mounting

Locking drives via UUID ensures they never mount to the OS NVMe if a dock reconnects.

1. Get the UUID

```
Bash  
sudo blkid  
# Copy the UUID for the raw partition (e.g., /dev/sdX1)
```

2. Configure /etc/crypttab (Unlocking)

Note: Never use the nofail flag in this file. It causes boot errors.

```
Bash  
sudo nano /etc/crypttab  
# Add this line:  
newdrive_vault UUID=YOUR-UUID-HERE /root/.keys/newdrive.key luks,discard
```

3. Configure /etc/fstab (Mounting)

Note: nofail belongs here so the Pi boots even if the drive is unplugged.

```
Bash  
sudo mkdir -p /mnt/newdrive_storage
```

```
sudo nano /etc/fstab
# Add this line:
/dev/mapper/newdrive_vault /mnt/newdrive_storage ext4 defaults,nofail 0 2
```

4. Apply and Verify

```
Bash
sudo systemctl daemon-reload
sudo mount -a
df -h
```

Phase 3: Hardware Armor (Orico Dock & SMR Drive Optimizations)

Essential for preventing JMicron chip lockups and SMR cache exhaustion during massive 40GB+ transfers.

1. Apply the 180-Second SCSI Timeout Rule

Prevents the Linux kernel from dropping the drive when the SMR heads stall to repair bad sectors.

```
Bash
echo 'ACTION=="add", SUBSYSTEM=="scsi", ENV{DEVTYPE}=="scsi_device",
ATTR{device/timeout}="180"' | sudo tee /etc/udev/rules.d/99-usb-timeout.rules
sudo udevadm control --reload-rules
sudo udevadm trigger
```

2. Apply the 15MB/s Write Straitjacket (cgroups)

Prevents the JMicron USB chip from overheating by throttling sequential write blasts.

```
Bash
# Convert the mount path to a systemd unit name (e.g., mnt-newdrive_storage.mount)
sudo systemctl set-property mnt-newdrive_storage.mount
```

IOWriteBandwidthMax=/dev/mapper/newdrive_vault 15M

Phase 4: Ultimate Permissions (Samba + Nextcloud Synergy)

Unifies read/write/delete access so Windows File Explorer and Docker Nextcloud do not fight over directory ownership.

1. Make the Users Share Groups

Bash
sudo usermod -aG www-data redwannabil
sudo usermod -aG redwannabil www-data

2. Force Co-Ownership and 775 Permissions

Bash
Assign the owner to redwannabil and the group to www-data
sudo chown -R redwannabil:www-data /mnt/newdrive_storage
Grant full Read(4)+Write(2)+Execute/Delete(1) to both Owner and Group (77)
sudo chmod -R 775 /mnt/newdrive_storage

Phase 5: Network Mapping

1. Add to Samba (/etc/samba/smb.conf)

Create an explicit block. Do not browse through PiMaster in Windows.

Ini, TOML
[New_Drive]
comment = New Encrypted Storage
path = /mnt/newdrive_storage
writeable = yes
force user = redwannabil
browseable = yes

guest ok = no
create mask = 0777
directory mask = 0777

Restart Samba: `sudo systemctl restart smbd nmbd`

2. Add to Nextcloud (Docker)

In the Nextcloud Web UI, go to **Administrative Settings -> External Storage**.

- **Folder Name:** New_Drive
- **External Storage Type:** Local
- **Configuration (Path):** /mnt/newdrive_storage
- Click the checkmark to save.

Phase 6: Monitoring Diagnostics

Watch real-time hardware performance in SSH/PowerShell to verify files are writing to the physical disks (wMB/s) and not leaking to the OS NVMe (nvme0n1).

Bash

Refreshes every 2 seconds in Megabytes
`iostat -x -m 2`

- **wMB/s:** Shows actual write speed (Watch for the 15MB/s limit).
- **w_await:** If this spikes over 1500ms, the SMR drive has hit a bad sector and is repairing itself. The 180s timeout rule is actively keeping it alive.

Phase 7: Emergency Recovery & Maintenance

If a dying SMR drive (like cctv_hdd) drops offline, causes an Input/output error, or gets locked in a Read-only file system state, run this block to wake it up instantly.

The "Wake-Up" Script:

Bash

1. Force unmount the stuck directory

```
sudo umount -l /mnt/cctv_hdd
```

2. Close the broken encryption channel

```
sudo cryptsetup luksClose cctv_hdd
```

3. Restart the crypto disk from /etc/crypttab

```
sudo cryptdisks_start cctv_hdd
```

4. Remount all drives from /etc/fstab

```
sudo mount -a
```

(Replace cctv_hdd with the specific drive name if a different drive crashes).

The Nextcloud Synchronization Command:

Anytime you manually copy files into a drive via Windows/Samba, Nextcloud's database won't know they exist. Run this exact Docker command to force Nextcloud to rescan everything and update the web UI:

Bash

```
sudo docker exec --user www-data nextcloud-app-1 php occ files:scan --all
```